Objective

To conduct research into, and create an autonomous driving system in Python to teach a GT3 race car to map out and drive itself around a track, improving its line and lap times over repeated laps using only telemetry data in the high-fidelity racing simulator *Le Mans Ultimate*.

Materials Covered / Justification

The primary resources for this project will be the articles *Learning How to Autonomously Race a Car: a Predictive Control Approach*[1], *Autonomous Racing using Learning Model Predictive Control*[2], and *DeepRacing: Parameterized Trajectories for Autonomous Racing*[3]. The first two articles focus on developing autonomous race cars that use data from previous laps to improve future performance, while *DeepRacing* focuses on autonomous driving and trajectory planning inside of a commercial racing simulator. These provide valuable insight and guidance for creating a similar system in *Le Mans Ultimate*, where the car would first map out a track at low speed, then use the collected telemetry to begin driving racing-speed laps and improve its driving as more data is collected.

Course Outline / Milestones

The course and development would be split into four major sections:

- Setup and initial track mapping (Setting up the test environment and tools, giving the program access to the telemetry and controls needed to drive, and completing an initial lap at pit speed to collect enough data to map out the circuit)
- Implement input methods and software interpretations of the race car and track (Creating a way for the program to interpret the telemetry data, determine where the car is on the track, and create an initial line and speed profile based on the shape of the circuit and the car's properties)
- Baseline autonomous driving (Creating full-length racing-speed laps around a circuit, while collecting and storing telemetry from each lap to build a more detailed understanding of the track and the limits of the car)
- Advanced lap improvement (Using data from previous laps to make adjustments to the racing line, braking points, speed, steering, throttle, and braking to progressively improve lap times)

Outcome / Assessments

Completion and understanding of the topic should result in an autonomous GT3 car being able to map out a circuit from a low-speed lap, then complete full racing-speed laps and improve its lap times as it collects more data. The final goal will be a lap time within 15% of a human-driven control lap time. Assessments and indications of progress include graphs showing lap time improvement over repeated laps, as well as onboard replays, racing line comparisons, and telemetry from the fastest autonomous lap.